Lewy Body Dementia

Samantha K. Holden, Zachary Trottier, and Morgan Farley, University of Colorado, Department of Neurology, Behavioral Neurology Section, Aurora, CO, United States

Key Points

- Lewy body diseases, characterized neuropathologically by alpha-synuclein and Lewy body deposition, include Parkinson's disease, Parkinson's disease dementia, and dementia with Lewy bodies.
- Lewy body dementia is an umbrella term that encompasses both Parkinson's disease dementia and dementia with Lewy bodies.
- Parkinson's disease dementia and dementia with Lewy bodies are primarily distinguished by the relative timing of onset and prominence of motor versus cognitive features.
- Core clinical features of dementia with Lewy bodies include parkinsonism, visual hallucinations, REM sleep behavior disorders, and cognitive fluctuations.
- Treatment of Lewy body dementia should focus on bothersome, quality-of-life limiting, or unsafe symptoms rather than following a standardized treatment algorithm, although a trial with a cholinesterase inhibitor is recommended for most patients.

Abstract

Lewy body dementia encompasses a spectrum of neurodegenerative disease causing progressive cognitive impairment. The disease entities included under a Lewy body dementia umbrella are dementia with Lewy bodies and Parkinson's disease dementia. These two conditions are currently best distinguished by the timing of onset and relative prominence of motor versus cognitive symptoms, yet they share many clinical and pathological characteristics. Criteria have also been developed or proposed to identify their respective prodromal stages, including mild cognitive impairment in Parkinson's disease and prodromal dementia with Lewy bodies. This article will overview the spectrum, clinical features, pathophysiologic mechanisms, diagnostic criteria, and management of LBD.

Overview of Lewy Body Dementia

Lewy body dementia (LBD) describes a spectrum of neurodegenerative diseases characterized by progressive cognitive impairment that coincides with parkinsonism. The defined disease entities under the LBD umbrella are dementia with Lewy bodies (DLB) and Parkinson's disease dementia (PDD). DLB is generally considered a primary cognitive syndrome with less prominent motor features, whereas PDD is inversely a primary motor disease with cognitive sequelae later in the disease course (Table 1). Both share cognitive and behavioral features associated with Lewy body pathology, including fluctuating arousal and awareness, well-formed visual hallucinations, and REM sleep disturbances. Parkinsonian motor symptoms in LBD can include bradykinesia, resting tremor, rigidity, gait impairment, and postural instability. The primary distinguishing factor between DLB and PDD is the timing of

 https://doi.org/10.1016/B978-0-323-95702-1.00129-9

Table 1 Distinguishing features of Parkinson's disease dementia from dementia with Lewy bodies.

	Overlap	*More DLB-specific*	*More PDD-specific*
Clinical features	Rigidity, bradykinesia	Less resting tremor, can be more postural and/or action tremor	Prominent resting tremor
	Non-motor symptoms (anosmia, constipation, apathy, REM sleep behavior disorder, autonomic dysfunction)	Earlier onset of dementia (same time or earlier than motor symptoms)	Later onset of dementia (>1 year from motor onset)
	Hallucinations (most commonly visual)	Well-formed visual hallucinations with impaired insight	Insight more maintained, more subtle phenomena (presence, passage hallucinations or visual illusions)
Neuropsychological profile	Executive and visuospatial dysfunction	More impaired attention and episodic verbal memory	Spared language and less impaired episodic memory
		More rapid cognitive decline	Usually slow, gradual cognitive decline
		Recurrent delirium	
Treatment response	Good response to cholinesterase inhibitors	Less levodopa-responsive	Levodopa-responsive
	Neuroleptic sensitivity	Degree of parkinsonism may not warrant dopaminergic therapy at all	Motor fluctuations (dyskinesias, ON-OFF phenomenon) with dopaminergic therapies
Imaging findings	Abnormal dopamine transporter (DaT) scan	More diffuse cortical atrophy	No specific findings
	Maintained mesial temporal volumes on structural MRI brain		
	Posterior hypometabolism on FDG-PET		
Neuropathological features	Alpha-synuclein pathology	Lewy body pathology in cortex and limbic structures	Lewy body pathology in cortex, limbic structures, and brainstem
		More prominent hippocampal Lewy body deposition	More neuronal loss in substantia nigra
		More AD co-pathology (up to 80%)	Less AD co-pathology (about 30%)

symptom onset, for which the "1-year rule" is generally applied. That is, if motor symptoms precede cognitive decline by at least 1 year the diagnosis is more likely PDD. If cognitive symptoms precede or emerge contemporaneously with the motor symptoms, then DLB is the likely diagnosis. Given that there is a shared pathologic underpinning to both diseases—alpha-synuclein deposits in the form of Lewy bodies—there is considerable debate as to whether these are truly distinct disease entities, or rather different forms of the same disease.

LBD is the second most common neurodegenerative etiology of cognitive decline, behind only Alzheimer's disease (AD). Although the true figure is unknown, meta-analyses suggest that the proportion of dementia attributable to DLB may be as high as 22.8% (Vann Jones and O'Brien, 2014). Incidence of DLB and PDD combined is estimated to be 5.9 per 100,000 person-years, a figure that is higher in men and markedly increases with age (Savica et al., 2013). An estimated 3%–4% of all dementia cases may be due to PDD specifically (Aarsland et al., 2005), and among individuals with PD the point prevalence of dementia is estimated to be 24%–31% (Hanagasi et al., 2017). As criteria for mild cognitive impairment secondary to DLB or PDD have been proposed more recently, there are few data regarding their respective prevalence figures. As with other neurodegenerative diseases, the prevalence of LBD is increasing due to a combination of globally increasing lifespans, increased prevalence of risk factors among elderly populations, as well as advances in diagnostic sensitivity.

LBD is thought to have a more rapid rate of cognitive and functional decline compared to other causes of dementia, with studies estimating an average 4-year survival after diagnosis with DLB, relative to the average survival of 5–7 years after diagnosis with AD. Clinical studies have noted that patients with LBD, as compared to those with AD, exhibit more rapid functional declines, convey greater caregiver burden, and are greater utilizers of healthcare resources (Kramberger et al., 2017). To accurately interpret these data, the impact of common delays to diagnosis in LBD, in addition to our limited understanding of disease staging, must be considered. LBD is often misdiagnosed, most frequently as AD or Parkinson's disease (PD), depending on the relative severity of cognitive or motor symptoms. While this problem has improved with ongoing revision of diagnostic criteria, diagnostic sensitivity remains relatively low. Despite the established variety of causes of dementia, the nuances that distinguish different etiologies of cognitive decline can puzzle even skilled diagnosticians. Thus, it is imperative that individuals experiencing cognitive decline undergo an adequate workup such that patients, caregivers, and family members can be accurately informed regarding treatment options, prognosis, and expectations for the disease course, as well as provided opportunities for inclusion in clinical trials.

Pathophysiology

Molecular Pathogenesis

The relationship between Lewy body deposition and cognitive function has been a source of inquiry since their initial observation in the early 20th century. Lewy bodies were initially observed in the brains of individuals diagnosed with "paralysis agitans," the disease that would come to be known as PD. Microscopy at the time recognized intracellular inclusion bodies distributed throughout the CNS, notably in the brainstem nuclei and throughout the cortex. It was not until the mid-late 20th century that the clinical syndrome at the intersection of PD and dementia was well-described. Alpha-synuclein was determined to be the primary component of Lewy body deposits in the 1990s, and around that time LDB came to be recognized as a distinct pathologic entity.

Alpha-synuclein is a protein encoded by the *SNCA* gene on chromosome 4q21 (Jakes et al., 1994). SNCA is expressed widely throughout the human central nervous system and is believed to play a role in regulating synaptic function and plasticity (Lashuel et al., 2013). Certain *SNCA* mutations have been identified as causing Lewy body pathology. Other genes implicated based on genome-wide association studies include *GBA* (encoding beta glucocerebrosidase) and *APOE4*, the best characterized risk gene for AD. These genes appear to play a small collective role, with most cases being without clear genetic linkage yet.

On a cellular level, it appears that misfolding and progressive accumulation of alpha-synuclein eventually leads to deposits observed as Lewy bodies. When deposited near synaptic terminals, vesicular trafficking and neurotransmitter release is impaired, ultimately resulting in the loss of dendritic spines and subsequent neuronal degeneration and cell loss (Schulz-Schaeffer, 2015). This may be the mechanism by which Lewy bodies alter synaptic function.

Neuropathological Distribution

LBD is known to present a wide variety of neuropathological topography. Primary neuropathological findings include degeneration of dopaminergic neurons in the substantia nigra, the neocortex, and various loci in the brainstem. There appear to be different patterns of distribution appreciated in LBD, characterized as either brainstem predominant, limbic, olfactory, or diffuse neocortical (McKeith et al., 2017). Of these subtypes, the diffuse neocortical and limbic appear to be most likely to produce the defined clinical syndrome of DLB. The Braak staging system, which was developed to categorize PD pathology, theorizes that pathology begins peripherally and gradually involves the brainstem, only reaching the cortex in severe cases (Braak et al., 2003). While this is suggestive of a potential pattern for all synucleinopathies, it is likely not wholly representative of distribution in DLB.

While the identification of pathogenic *SNCA* mutations lends credence to the notion that alpha-synuclein is a primary driver of LBD pathology, it is certainly not the only factor at play. Cortical alpha-synuclein seems to correlate well with severity of cognitive decline and be more tightly related to accuracy of DLB diagnosis; however, beta-amyloid and tau deposits are also seen in many patients with DLB and PDD. Some research suggests interactions and potential synergistic effects between neurodegenerative pathologies, with the likelihood of co-pathologies increasing with age. Furthermore, there are likely contributions of cerebrovascular and neuroinflammatory pathology to LBD, which remain to be better elucidated.

Functional Implications of Neuropathology

Cognitive decline in LBD follows a trajectory unique from other causes of dementia. DLB is characterized by prominent fluctuations in awareness and attention, along with visuospatial and executive dysfunction, which are then followed by amnestic changes later in the disease course when compared to AD. In PDD, even though cognitive symptoms are later sequelae of PD, they do appear to have a similar pattern of decline once present. It is believed that the cognitive symptoms of LBD are largely related to cholinergic and dopaminergic degeneration in subcortical structures. These same changes are thought to contribute to the neuropsychiatric sequelae of LBD, which include hallucinations, delusions, depression, anxiety, and apathy. The established occipital cortical pathology is thought to underlie the visual hallucinations and visuospatial dysfunction seen in most patients, and this theory is consistent with functional imaging revealing hypoperfusion and hypometabolism of the occipital cortex (Outeiro et al., 2019).

REM sleep behavior disorder (RBD) is often one of the earliest, prodromal symptoms seen in patients with Lewy body disease. RBD may predate cognitive decline by as many as 10 years, making it one of the earliest indicators of an incipient synucleinopathy (Boeve et al., 2013). Studies suggest that the reticular formation, a collection of brainstem nuclei that plays a role in modulation of sleep cycles and daytime awareness, are among the earliest tissues affected by alpha-synuclein deposition (Ferreira et al., 2021).

Motor symptoms are characterized by parkinsonism, including bradykinesia, rigidity, postural instability, and resting tremor. These changes are likely resultant from the degenerative changes in the substantia nigra and striatum. Parkinsonian features are observed in about 85% of patients with DLB (McKeith et al., 2017). While parkinsonism is itself the central feature of PD, the motor changes observed occur later in DLB and provide diagnostic distinction through their sequential chronicity.

Autonomic symptoms are poorly understood at present, but working theories suggest that the disproportionate brainstem involvement unique to synucleinopathies is the primary driver. Furthermore, some studies suggest that alpha-synuclein may affect aspects of organelle transport, thus affecting CNS neurons in a length-dependent manner (Volpicelli-Daley et al., 2014). This insight may offer explanation as to the mechanism of cardiac sympathetic denervation, which is an established early biomarker for PD, PDD, and DLB.

Diagnosis of Lewy Body Dementia

As our understanding of the disease processes driving LBD has developed over recent decades, the framework with which we have diagnosed the disease has likewise been refined over time. Understanding the breadth of the spectrum of LBD has allowed for more accurate diagnosis of related diseases as well. In addition to improved understanding of clinical signs and symptoms, novel biomarkers have greatly accelerated the improvement in diagnostic specificity. These advances are much needed, as they allow for more precisely directed treatment regimens and more accurate prognoses at the time of diagnoses.

Dementia With Lewy Bodies

The fourth and most recent consensus report on the diagnostic criteria for DLB was developed in 2017 (McKeith et al., 2017). Firstly, a patient is required to demonstrate signs of dementia for the diagnosis to be valid; the symptoms must be of great enough severity that they interfere with activities of daily living. The remainder of the diagnosis relies on the presence of several core and supportive clinical features, with the aid of indicative and supportive biomarkers. The four core clinical features of DLB are: (1) fluctuating cognition; (2) recurrent visual hallucinations; (3) REM sleep behavior disorder; and (4) one or more cardinal features of parkinsonism. Supportive features include sensitivity to antipsychotic medications, postural instability, repeated falls, syncope, autonomic dysfunction, hyposmia, delusions, and other neuropsychiatric sequelae. Indicative biomarkers include: (1) reduced dopamine transporter uptake in basal ganglia demonstrated by SPECT or PET; (2) low uptake on ^{123}iodine-MIGB myocardial scintigraphy; and (3) polysomnography showing REM sleep without atonia. Other biomarkers may also be considered supportive and will be discussed in detail in the below **Diagnostic Biomarkers** section.

According to the above criteria, the diagnosis of DLB is categorized as probable DLB or possible DLB. Probable DLB is diagnosed when: (1) two or more core clinical features are present; or (2) one core clinical feature is present as well as one indicative biomarker. Possible DLB is diagnosed if: (1) one core clinical feature is present without biomarkers; or (2) one or more indicative biomarkers are present without core clinical features. Compared to the previous consensus report published in 2005 (McKeith et al., 2005), these diagnostics rely more heavily on biomarker use and now consider REM sleep behavior disorder to be a core clinical feature. One common conundrum that clinicians face in the diagnosis of DLB is that when patients present late in disease course, they often have both motor and cognitive symptoms of such severity that it can be difficult to elicit which was the initial symptom. In these scenarios, a careful clinical history focused on the chronology of symptom onset is crucial for an accurate diagnosis.

Prodromal Dementia With Lewy Bodies

The term *prodromal DLB* describes several onset patterns of early Lewy body disease which may progress to DLB: (1) MCI with Lewy bodies (MCI-LB); (2) delirium-onset prodromal DLB; and (3) psychiatric-onset prodromal DLB. Diagnostic criteria have only recently been developed for MCI-LB (McKeith et al., 2020) and are proposed for research use only currently. Prerequisites for the diagnosis of prodromal DLB include concern for cognitive decline (from patient, caregiver, or clinician), objective evidence of impairment in a cognitive domain, and maintained performance of activities of daily living (to distinguish from dementia-level severity). Core clinical features are as described above for DLB, as are the proposed biomarkers. The diagnostic difference distinguishing MCI-LB and DLB lies mostly in the patient's level of functioning. As with probable DLB, probable MCI-LB is diagnosed if: (1) two or more core clinical features are present; or (2) one or more core clinical features is present with one or more proposed biomarkers. Similarly, possible MCI-LB is diagnosed if: (1) only one core clinical feature is present; or (2) one or more biomarkers is present without core clinical features. The supportive clinical features overlap with those in the above DLB criteria, with a single addition of prolonged or recurrent delirium. Cognitive profiles may be helpful in distinguishing between MCI-LB and MCI-AD; non-amnestic MCI profiles are 10−15 times more likely to progress to LBD compared to amnestic MCI (which are more likely to progress to AD) (Sadiq et al., 2017). While there are not yet established criteria for psychiatric-onset or delirium-onset DLB, these are emerging areas which may soon have criteria of their own for diagnosis.

Parkinson's Disease Dementia

The diagnostic criteria for PDD were established in 2007 (Emre et al., 2007). Diagnosis of PDD requires a pre-existing diagnosis of PD, objective impairment on cognitive testing (with more than one cognitive domain affected), and impaired activities of daily living related to cognitive rather than motoric dysfunction. Given the impact of parkinsonian motor symptoms like tremor and bradykinesia on daily function, including basic activities of daily living like dressing and grooming, careful history from the patient and a knowledgeable informant are required to distinguish cognitive from motoric functional impairment. Cognitive features include fluctuating attention, executive dysfunction, visuospatial dysfunction, and impaired free recall. Behavioral features, such as apathy, excessive daytime sleepiness, and psychosis, may be supportive of a PDD diagnosis, but are not required. As with other dementias, other dementia etiologies, including vascular dementia, systemic disease, drug intoxication, and/or major depression, must be excluded prior to making the diagnosis of PDD.

Mild Cognitive Impairment in Parkinson's Disease

The most widely accepted diagnostic criteria for MCI-PD were published in 2012 (Litvan et al., 2012). In this framework, PD-MCI can be diagnosed by Level I (abbreviated assessment) or Level II (comprehensive assessment) criteria. Level II PD-MCI diagnosis requires evaluation of at least two tests in each of the five cognitive domains (memory, attention, executive, visuospatial, and language), with objective impairment on two or more cognitive tests. PD-MCI can be classified as single- or multi-domain, depending on the number of cognitive domains in which impairment is observed. A diagnosis of PD-MCI requires determination of intact daily function, which is the major separating feature between MCI and dementia level cognitive impairment. Patients with PD-MCI convert to PDD at a rate of approximately 10%−15% per year.

Diagnostic Biomarkers

Historically, LBD has been a diagnosis primarily made using clinical and historical information. In recent years, a variety of useful biomarkers have been developed that have aided in diagnosis as well as provided key insights into pathophysiologic mechanisms of the synucleinopathies. Biomarkers currently used in the diagnostic evaluation of suspected LBD include structural brain MRI, dopamine transporter scans (SPECT, PET), electroencephalogram (EEG), polysomnography, and cardiac scintigraphy. Structural neuroimaging is often used to rule out other etiologies of cognitive decline, such as AD or vascular dementia, rather than positively identifying LBD. In DLB, MRI imaging can show globally decreased cortical gray matter volumes with a relative sparing of the mesial temporal lobe as compared to patients with AD. In PDD, the sparing of the mesial temporal lobes is also present but there tends to be less cortical atrophy than in DLB or AD. Molecular imaging with fluorodeoxyglucose-positron emission tomography (FDG-PET) scans in LBD can show reduced posterior (occipital and parietal) metabolism, which can fit with cortical visual symptomatology. The "cingulate island sign," or sparing of the posterior cingulate cortex from the posterior hypometabolism pattern on FDG-PET, can help distinguish LBD from AD (Imabayashi et al., 2017).

Dopamine transporter (DAT) imaging has emerged as a sensitive and specific modality for detection of primary parkinsonian disorders, including PDD and DLB. DAT is a membrane protein primarily expressed in the presynaptic membranes of dopaminergic neurons, most highly concentrated in the striatum. A reduction in DAT binding in the striatum has been shown to be of high sensitivity (77.7%) and specificity (90.4%) for diagnosing synucleinopathies (McKeith et al., 2007), though it cannot distinguish primary parkinsonian disorders from one other.

^{123}Iodine-metaiodobenzylguanidine (MIBG) myocardial scintigraphy is another biomarker that has proven valuable in the identification of synucleinopathies. MIBG is a radioisotope that binds to presynaptic cardiac autonomic nerve terminals. It has been well-established that cardiac autonomic denervation occurs in PD and DLB, thus MIGB uptake is reduced in both conditions.

Studies evaluating the diagnostic accuracy of MIGB uptake scans have found a sensitivity of 69.7% and a specificity of 87.0% (Yoshita et al., 2015). This high specificity provides excellent utility in discriminating between synucleinopathies and non-synucleinopathies. Furthermore, more recent studies have shown that the specificity may remain high even in LBD patients in the MCI stage, suggesting that these tests are likely useful even early in the disease course. It is worth noting, however, that comorbid cardiac disease or medication use can confound these results and it is not currently FDA-approved for this indication in the United States.

Polysomnography has been used for some time, but its diagnostic role has grown in recent years with the increased recognition as RBD as a core feature of LBD. Standard protocol for polysomnographic evaluation includes measurement of EEG activity, monitoring eye movement, and monitoring motor activity, among other factors during sleep. The most pertinent finding on polysomnography is REM sleep without atonia, which has a >90% predictive value for diagnosis of an underlying synucleinopathy (Boeve et al., 2013).

EEG can be helpful in the diagnosis of DLB, particularly in distinguishing between synucleinopathies and non-synucleinopathies. Particularly, studies comparing DLB and AD have shown that patients with DLB have lower peak frequencies, greater slow wave activity, and more prominent connectivity disruptions (van der Zande et al., 2018). Quantitative EEG patterns may be useful in predicting the transformation from MCI-LB to DLB and may thus have value in prognosis of early-stage disease (Bonanni et al., 2015) and are included as supportive biomarkers in the proposed prodromal DLB criteria.

Numerous fluid and tissue samples have been proposed as biomarkers to aid in the diagnosis of LBD. Multiple groups have reported elevated alpha-synuclein noted in skin biopsies in PD, but it is unclear at which disease stage this occurs in LBD. In addition, CSF profiles are undergoing investigation as a discriminatory test in LBD, including alpha-synuclein seeding assays using real-time quaking-induced conversion (RT-QuIC) (Grossauer et al., 2023). A recent consensus report notes that while there is great potential for biomarkers in the near future, none of these are yet ready for clinical application until they are better characterized (Scott et al., 2021).

Management of Lewy Body Dementia

There are currently no curative or disease-modifying treatment options available for LBD. Current treatment options revolve primarily around managing cognitive, behavioral, and motor symptoms. Adequate treatment is complicated by the poor diagnostic precision as well as the wide variability in symptoms observed on an individual basis. The unique cognitive and behavioral profiles in combination with the motor symptoms of LBD necessitate unique treatment strategies.

The primary cognitive features in LBD include fluctuating cognition as well as impaired attention, executive, and visuospatial functions. The best studied medications in DLB are the cholinesterase inhibitors, namely donepezil and rivastigmine. Studies have shown similar levels of effect for cognitive improvement in both DLB and PDD, which is often a more robust response than seen in other forms of dementia, including AD. Furthermore, cholinesterase inhibitors can reduce caregiver burden and treat bothersome hallucinations and delusions. Thus, cholinesterase inhibitors are recommended as first line treatment options for cognitive symptoms of LBD, though only rivastigmine is FDA-approved for use in PDD. Patients with LBD often demonstrate a more noticeable clinical response to cholinesterase inhibitors than patients with AD, both for benefits at initiation as well as detriments with withdrawal, especially if sudden. Memantine, an NMDA receptor antagonist, has been studied to some extent yielding mixed results for efficacy though is generally well-tolerated. An open dialog regarding expectations with patients Is important when considering cholinesterase inhibitor treatment, as these are medications meant to preserve current levels of function rather than increase function.

Neuropsychiatric sequelae in LBD can be highly variable, and thus a plethora of symptom-driven treatment options may be considered. Hallucinations are the most common sequelae and are a core clinical feature of DLB. While they can be distressing, many patients are not overly bothered by visual hallucinations or other sensory disturbances. While non-pharmacologic interventions such as environmental modification may be helpful, patients experiencing severe or distressing psychotic symptoms may require treatment. In these cases, a cholinesterase inhibitor is a safe and often effective first-line option for treatment of psychosis in LBD, if not already prescribed for cognitive benefit. Next, oral quetiapine is often the best tolerated antipsychotic therapy option and may confer moderate benefit (Takahashi et al., 2003). Other antipsychotic agents such as olanzapine and clozapine have been studied with mixed results and higher levels of sensitivity reactions, as well as careful monitoring required with clozapine (Taylor et al., 2020). Pimavanserin, a 5-HT2a inverse agonist, was approved for treatment of PD-associated psychosis in 2016, though not specifically approved for use in DLB at this time. Depression and anxiety are also quite common among patients with LBD, affecting up to one third of patients (Kuring et al., 2018). Meta-analyses of patients with PD and comorbid depression suggest that SSRIs, SNRIs, and tricyclics may all improve depressive and anxious symptoms without significantly worsening motor symptoms. In addition, cholinesterase inhibitors may also benefit mood symptoms in addition to global cognitive function. As with other medications, the preferred strategy is to slowly titrate to the lowest effective dose until benefit is reached. Behavioral interventions such as CBT as well as physiotherapy have also shown benefit in patients with mood symptoms diagnosed with PD but have been less studied in DLB specifically.

Although most patients with LBD experience motor symptoms to some extent, there is great heterogeneity in the severity and nature of those motor symptoms. Thus, optimizing treatment regimens for patients with LBD can be challenging. PDD itself presents a particular problem in that many patients with PDD have been diagnosed and treated for PD for many years, and therefore require more complicated dopaminergic regimens, often with motor fluctuations, with their dopaminergic therapies potentially exacerbating cognitive and behavioral symptoms. A careful balance between motor-focused therapies and cognitive and behavioral-focused therapies is required. In DLB, the degree of parkinsonism is often less severe than in PD and may not require dopaminergic therapy at all; if required, focusing on levodopa, rather than monoamine oxidase inhibitors or dopamine agonists, is recommended to better balance motor benefit with avoidance of side effects, including potentially worsening psychosis.

There are several treatments undergoing investigation for potential use in LBD. The recent approval of disease-modifying therapies in AD has intensified the pressure to pursue novel therapeutics for other forms of dementia, including LBD. Tyrosine kinase inhibitors (TKIs) are thought to promote autophagy and thus enhance clearance of alpha-synuclein, beta-amyloid, and phosphorylated tau (Fagiani et al., 2020). As of 2022, phase 2 clinical trials were underway for vodobatinib and nilotinib, both tyrosine kinase inhibitors. Bosutinib, another TKI showed good tolerability in addition to improved activities of daily living in DLB, however questions remain about its potential long-term impact on the disease course (Pagan et al., 2022). CT1812 is another disease-modifying therapy undergoing investigation, which exerts effect by inhibiting alpha-synuclein oligomeric toxicity in addition to modulating autophagy and cellular trafficking (Limegrover et al., 2021). Given the known genetic association of glucocerebrosidase with LBD, ambroxol is a compound designed to enhance glucocerebrosidase activity and lysosomal function. These actions enhance autophagy and reduce alpha-synuclein levels (Balestrino and Schapira, 2018). There are also a variety of pharmacologic therapies targeting neurotransmitter systems under investigation at present, which may also aid in symptom management.

Conclusions and Future Directions

As the second most common neurodegenerative etiology of dementia, LBD deserves increased focus, research funding, and professional and community education. LBD is the costliest form of dementia, with the highest degree of caregiver burden, and the most delayed in accurate diagnosis. Precision and accuracy of terminology, classification, and staging will be needed to make meaningful improvements in LBD detection and treatment, as well as eventual disease modification and prevention. The past few decades have seen significant advancement in our understanding of the disease process driving LBD, and consequently its diagnosis and management. Improved diagnostic recognition has led to greater awareness, enhanced research of the pathophysiology involved, and thus new biomarkers. Although there remain no disease modifying treatments for LBD, strides in our understanding of the disease course have allowed for targeted symptomatic management, albeit with modest effects overall. Further understanding of the underlying genetics, molecular mechanisms of disease, and the neuronal networks involved will prove critical in future development of potential disease-modifying therapies.

References

Aarsland, D., Zaccai, J., Brayne, C., 2005. A systematic review of prevalence studies of dementia in Parkinson's disease. Mov. Disord. 20, 1255–1263.

Balestrino, R., Schapira, A.H.V., 2018. Glucocerebrosidase and Parkinson disease: molecular, clinical, and therapeutic implications. Neuroscientist 24, 540–559.

Boeve, B.F., Silber, M.H., Ferman, T.J., Lin, S.C., Benarroch, E.E., Schmeichel, A.M., Ahlskog, J.E., Caselli, R.J., Jacobson, S., Sabbagh, M., Adler, C., Woodruff, B., Beach, T.G., Iranzo, A., Gelpi, E., Santamaria, J., Tolosa, E., Singer, C., Mash, D.C., Luca, C., Arnulf, I., Duyckaerts, C., Schenck, C.H., Mahowald, M.W., Dauvilliers, Y., Graff-Radford, N.R., Wszolek, Z.K., Parisi, J.E., Dugger, B., Murray, M.E., Dickson, D.W., 2013. Clinicopathologic correlations in 172 cases of rapid eye movement sleep behavior disorder with or without a coexisting neurologic disorder. Sleep Med. 14, 754–762.

Bonanni, L., Perfetti, B., Bifolchetti, S., Taylor, J.P., Franciotti, R., Parnetti, L., Thomas, A., Onofrj, M., 2015. Quantitative electroencephalogram utility in predicting conversion of mild cognitive impairment to dementia with Lewy bodies. Neurobiol. Aging 36, 434–445.

Braak, H., Del Tredici, K., Rub, U., De Vos, R.A., Jansen Steur, E.N., Braak, E., 2003. Staging of brain pathology related to sporadic Parkinson's disease. Neurobiol. Aging 24, 197–211.

Emre, M., Aarsland, D., Brown, R., Burn, D.J., Duyckaerts, C., Mizuno, Y., Broe, G.A., Cummings, J., Dickson, D.W., Gauthier, S., Goldman, J., Goetz, C., Korczyn, A., Lees, A., Levy, R., Litvan, I., Mckeith, I., Olanow, W., Poewe, W., Quinn, N., Sampaio, C., Tolosa, E., Dubois, B., 2007. Clinical diagnostic criteria for dementia associated with Parkinson's disease. Mov. Disord. 22, 1689–1707 quiz 1837.

Fagiani, F., Lanni, C., Racchi, M., Govoni, S., 2020. Targeting dementias through cancer kinases inhibition. Alzheimers Dement. 6, e12044.

Ferreira, N., Richner, M., Van Der Laan, A., Bergholdt Jul Christiansen, I., Vaegter, C.B., Nyengaard, J.R., Halliday, G.M., Weiss, J., Giasson, B.I., Mackenzie, I.R., Jensen, P.H., Jan, A., 2021. Prodromal neuroinvasion of pathological alpha-synuclein in brainstem reticular nuclei and white matter lesions in a model of alpha-synucleinopathy. Brain Commun. 3, fcab104.

Grossauer, A., Hemicker, G., Krismer, F., Peball, M., Djamshidian, A., Poewe, W., Seppi, K., Heim, B., 2023. α-Synuclein seed amplification assays in the diagnosis of synucleinopathies using cerebrospinal fluid-A systematic review and meta-analysis. Mov. Disord. Clin. Pract. 10 (5), 737–747.

Hanagasi, H.A., Tufekcioglu, Z., Emre, M., 2017. Dementia in Parkinson's disease. J. Neurol. Sci. 374, 26–31.

Imabayashi, E., Soma, T., Sone, D., Tsukamoto, T., Kimura, Y., Sato, N., Murata, M., Matsuda, H., 2017. Validation of the cingulate island sign with optimized ratios for discriminating dementia with Lewy bodies from Alzheimer's disease using brain perfusion SPECT. Ann. Nucl. Med. 31, 536–543.

Jakes, R., Spillantini, M.G., Goedert, M., 1994. Identification of two distinct synucleins from human brain. FEBS Lett. 345, 27–32.

Kramberger, M.G., Auestad, B., Garcia-Ptacek, S., Abdelnour, C., Olmo, J.G., Walker, Z., Lemstra, A.W., Londos, E., Blanc, F., Bonanni, L., Mckeith, I., Winblad, B., De Jong, F.J., Nobili, F., Stefanova, E., Petrova, M., Falup-Pecurariu, C., Rektorova, I., Bostantjopoulou, S., Biundo, R., Weintraub, D., Aarsland, D., E-DLB, 2017. Long-term cognitive decline in dementia with Lewy bodies in a large multicenter, international cohort. J. Alzheimers Dis. 57, 787–795.

Kuring, J.K., Mathias, J.L., Ward, L., 2018. Prevalence of depression, anxiety and PTSD in people with dementia: a systematic review and meta-analysis. Neuropsychol. Rev. 28, 393–416.

Lashuel, H.A., Overk, C.R., Oueslati, A., Masliah, E., 2013. The many faces of alpha-synuclein: from structure and toxicity to therapeutic target. Nat. Rev. Neurosci. 14, 38–48.

Limegrover, C.S., Yurko, R., Izzo, N.J., Labarbera, K.M., Rehak, C., Look, G., Rishton, G., Safferstein, H., Catalano, S.M., 2021. Sigma-2 receptor antagonists rescue neuronal dysfunction induced by Parkinson's patient brain-derived alpha-synuclein. J. Neurosci. Res. 99, 1161–1176.

Litvan, I., Goldman, J.G., Troster, A.I., Schmand, B.A., Weintraub, D., Petersen, R.C., Mollenhauer, B., Adler, C.H., Marder, K., Williams-Gray, C.H., Aarsland, D., Kulisevsky, J., Rodriguez-Oroz, M.C., Burn, D.J., Barker, R.A., Emre, M., 2012. Diagnostic criteria for mild cognitive impairment in Parkinson's disease: movement Disorder Society Task Force guidelines. Mov. Disord. 27, 349—356.

Mckeith, I., O'brien, J., Walker, Z., Tatsch, K., Booij, J., Darcourt, J., Padovani, A., Giubbini, R., Bonuccelli, U., Volterrani, D., Holmes, C., Kemp, P., Tabet, N., Meyer, I., Reininger, C., DLB Study Group, 2007. Sensitivity and specificity of dopamine transporter imaging with 123I-FP-CIT SPECT in dementia with Lewy bodies: a phase III, multicentre study. Lancet Neurol. 6, 305—313.

Mckeith, I.G., Boeve, B.F., Dickson, D.W., Halliday, G., Taylor, J.P., Weintraub, D., Aarsland, D., Galvin, J., Attems, J., Ballard, C.G., Bayston, A., Beach, T.G., Blanc, F., Bohnen, N., Bonanni, L., Bras, J., Brundin, P., Burn, D., Chen-Plotkin, A., Duda, J.E., El-Agnaf, O., Feldman, H., Ferman, T.J., Ffytche, D., Fujishiro, H., Galasko, D., Goldman, J.G., Gomperts, S.N., Graff-Radford, N.R., Honig, L.S., Iranzo, A., Kantarci, K., Kaufer, D., Kukull, W., Lee, V.M.Y., Leverenz, J.B., Lewis, S., Lippa, C., Lunde, A., Masellis, M., Masliah, E., Mclean, P., Mollenhauer, B., Montine, T.J., Moreno, E., Mori, E., Murray, M., O'brien, J.T., Orimo, S., Postuma, R.B., Ramaswamy, S., Ross, O.A., Salmon, D.P., Singleton, A., Taylor, A., Thomas, A., Tiraboschi, P., Toledo, J.B., Trojanowski, J.Q., Tsuang, D., Walker, Z., Yamada, M., Kosaka, K., 2017. Diagnosis and management of dementia with Lewy bodies: fourth consensus report of the DLB Consortium. Neurology 89, 88—100.

Mckeith, I.G., Dickson, D.W., Lowe, J., Emre, M., O'brien, J.T., Feldman, H., Cummings, J., Duda, J.E., Lippa, C., Perry, E.K., Aarsland, D., Arai, H., Ballard, C.G., Boeve, B., Burn, D.J., Costa, D., Del Ser, T., Dubois, B., Galasko, D., Gauthier, S., Goetz, C.G., Gomez-Tortosa, E., Halliday, G., Hansen, L.A., Hardy, J., Iwatsubo, T., Kalaria, R.N., Kaufer, D., Kenny, R.A., Korczyn, A., Kosaka, K., Lee, V.M., Lees, A., Litvan, I., Londos, E., Lopez, O.L., Minoshima, S., Mizuno, Y., Molina, J.A., Mukaetova-Ladinska, E.B., Pasquier, F., Perry, R.H., Schulz, J.B., Trojanowski, J.Q., Yamada, M., Consortium on DLB, 2005. Diagnosis and management of dementia with Lewy bodies: third report of the DLB Consortium. Neurology 65, 1863—1872.

Mckeith, I.G., Ferman, T.J., Thomas, A.J., Blanc, F., Boeve, B.F., Fujishiro, H., Kantarci, K., Muscio, C., O'brien, J.T., Postuma, R.B., Aarsland, D., Ballard, C., Bonanni, L., Donaghy, P., Emre, M., Galvin, J.E., Galasko, D., Goldman, J.G., Gomperts, S.N., Honig, L.S., Ikeda, M., Leverenz, J.B., Lewis, S.J.G., Marder, K.S., Masellis, M., Salmon, D.P., Taylor, J.P., Tsuang, D.W., Walker, Z., Tiraboschi, P., Prodromal DLB Diagnostic Study Group, 2020. Research criteria for the diagnosis of prodromal dementia with Lewy bodies. Neurology 94, 743—755.

Outeiro, T.F., Koss, D.J., Erskine, D., Walker, L., Kurzawa-Akanbi, M., Burn, D., Donaghy, P., Morris, C., Taylor, J.P., Thomas, A., Attems, J., Mckeith, I., 2019. Dementia with Lewy bodies: an update and outlook. Mol. Neurodegener. 14, 5.

Pagan, F.L., Torres-Yaghi, Y., Hebron, M.L., Wilmarth, B., Turner, R.S., Matar, S., Ferrante, D., Ahn, J., Moussa, C., 2022. Safety, target engagement, and biomarker effects of bosutinib in dementia with Lewy bodies. Alzheimers Dement. 8, e12296.

Sadiq, D., Whitfield, T., Lee, L., Stevens, T., Costafreda, S., Walker, Z., 2017. Prodromal dementia with Lewy bodies and prodromal Alzheimer's disease: a comparison of the cognitive and clinical profiles. J. Alzheimers Dis. 58 (2), 463—470. PMID: 28453473. https://doi.org/10.3233/JAD-161089.

Savica, R., Grossardt, B.R., Bower, J.H., Boeve, B.F., Ahlskog, J.E., Rocca, W.A., 2013. Incidence of dementia with Lewy bodies and Parkinson disease dementia. JAMA Neurol. 70, 1396—1402.

Schulz-Schaeffer, W.J., 2015. Is cell death primary or secondary in the pathophysiology of Idiopathic Parkinson's disease? Biomolecules 5, 1467—1479.

Scott, G.D., Arnold, M.R., Beach, T.G., Gibbons, C.H., Kanthasamy, A.G., Lebovitz, R.M., Lemstra, A.W., Shaw, L.M., Teunissen, C.E., Zetterberg, H., Taylor, A.S., Graham, T.C., Boeve, B.F., Gomperts, S.N., Graff-Radford, N.R., Moussa, C., Poston, K.L., Rosenthal, L.S., Sabbagh, M.N., Walsh, R.R., Weber, M.T., Armstrong, M.J., Bang, J.A., Bozoki, A.C., Domoto-Reilly, K., Duda, J.E., Fleisher, J.E., Galasko, D.R., Galvin, J.E., Goldman, J.G., Holden, S.K., Honig, L.S., Huddleston, D.E., Leverenz, J.B., Litvan, I., Manning, C.A., Marder, K.S., Pantelyat, A.Y., Pelak, V.S., Scharre, D.W., Sha, S.J., Shill, H.A., Mari, Z., Quinn, J.F., Irwin, D.J., 2021. Fluid and tissue biomarkers of Lewy body dementia: report of an LBDA Symposium. Front. Neurol. 12, 805135.

Takahashi, H., Yoshida, K., Sugita, T., Higuchi, H., Shimizu, T., 2003. Quetiapine treatment of psychotic symptoms and aggressive behavior in patients with dementia with Lewy bodies: a case series. Prog. Neuro-Psychopharmacol. Biol. Psychiatr. 27, 549—553.

Taylor, J.P., McKeith, I.G., Burn, D.J., Boeve, B.F., Weintraub, D., Bamford, C., Allan, L.M., Thomas, A.J., O'Brien, J.T., 2020. New evidence on the management of Lewy body dementia. Lancet Neurol. 19, 157—169. https://doi.org/10.1016/S1474-4422(19)30153-X.

Van Der Zande, J.J., Gouw, A.A., Van Steenoven, I., Scheltens, P., Stam, C.J., Lemstra, A.W., 2018. EEG characteristics of dementia with Lewy bodies, Alzheimer's disease and mixed pathology. Front. Aging Neurosci. 10, 190.

Vann Jones, S.A., O'Brien, J.T., 2014. The prevalence and incidence of dementia with Lewy bodies: a systematic review of population and clinical studies. Psychol. Med. 44, 673—683.

Volpicelli-Daley, L.A., Gamble, K.L., Schultheiss, C.E., Riddle, D.M., West, A.B., Lee, V.M., 2014. Formation of alpha-synuclein Lewy neurite-like aggregates in axons impedes the transport of distinct endosomes. Mol. Biol. Cell 25, 4010—4023.

Yoshita, M., Arai, H., Arai, H., Arai, T., Asada, T., Fujishiro, H., Hanyu, H., Iizuka, O., Iseki, E., Kashihara, K., Kosaka, K., Maruno, H., Mizukami, K., Mizuno, Y., Mori, E., Nakajima, K., Nakamura, H., Nakano, S., Nakashima, K., Nishio, Y., Orimo, S., Samuraki, M., Takahashi, A., Taki, J., Tokuda, T., Urakami, K., Utsumi, K., Wada, K., Washimi, Y., Yamasaki, J., Yamashina, S., Yamada, M., 2015. Diagnostic accuracy of 123I-meta-iodobenzylguanidine myocardial scintigraphy in dementia with Lewy bodies: a multicenter study. PLoS One 10, e0120540.